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4.2 Practice Questions

Solve the problem and pick the best answer choice given.

Numbers: All numbers used are real numbers.

Figures: A figure in a Quantitative Reasoning question gives information useful in solving the problem. Figures are drawn as accurately as possible, except as noted. Lines shown as straight are straight. Lines that look jagged may also be straight. Points, angles, regions, etc., are positioned as shown. All figures are flat unless otherwise noted.

Questions 1 to 96 — Difficulty: Easy

1. In the graduating class of a certain college, 48 percent of the students identify exclusively as male and 52 percent identify exclusively as female. In this class, 40 percent of the students who identify as male and 20 percent of the students who identify as female are 25 years old or older. If one student in the graduating class is randomly selected, approximately what is the probability that the student will be less than 25 years old?

   A. 0.90

   B. 0.70

   C. 0.45

   D. 0.30

   E. 0.25

2. In a certain board game, a stack of 48 cards, 8 of which represent a single share of stock, are shuffled and then placed face down. If the first 2 cards selected do not represent shares of stock, what is the probability that the third card selected will represent a share of stock?

   A. $\frac{1}{8}$

   B. $\frac{1}{6}$

   C. $\frac{1}{5}$

   D. $\frac{3}{23}$

   E. $\frac{4}{23}$

3. During a trip that they took together, Carmen, Juan, Maria, and Rafael drove an average (arithmetic mean) of 80 miles each. Carmen drove 72 miles, Juan drove 78 miles, and Maria drove 83 miles. How many miles did Rafael drive?

   A. 80

   B. 82

   C. 85

   D. 87

   E. 89

4. Each week, a clothing salesperson receives a commission equal to 15 percent of the first $500 in sales and 20 percent of all additional sales that week. What commission would the salesperson receive on total sales for the week of $1,300?

   A. $195

   B. $227

   C. $235

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D. $260

E. $335

5. Five batches of 100 nails each are taken from a production line. The numbers of defective nails in the first four batches are 2, 4, 3, and 5, respectively. If the fifth batch has either 1, 2, or 6 defective nails, for which of these values does the average (arithmetic mean) number of defective nails per batch for the five batches equal the median number of defective nails for the five batches?

I. 1

II. 2

III. 6

A. I only

B. II only

C. III only

D. I and III only

E. I, II, and III

6. List S consists of 10 consecutive odd integers, and list T consists of 5 consecutive even integers. If the least integer in S is 7 more than the least integer in T, how much greater is the average (arithmetic mean) of the integers in S than the average of the integers in T ?

A. 2

B. 7

C. 8

D. 12

E. 22

Lane number	Number in line
1	5
2	4
3	3
4	4
5	2

7. A certain grocery store has 5 checkout lanes. The table above shows the amount of carts the last person waiting in each checkout lane had at closing time yesterday. What was the median number of carts?

A. $2\frac{4}{5}$

B. 3

C. $3\frac{1}{2}$

D. $3\frac{3}{5}$

E. 4

8. $\left(\frac{1}{2}-\frac{1}{3}\right)+\left(\frac{1}{3}-\frac{1}{4}\right)+\left(\frac{1}{4}-\frac{1}{5}\right)+\left(\frac{1}{5}-\frac{1}{6}\right)=$

A. $-\frac{1}{6}$

B. 0

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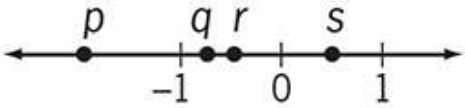
B. 0

C. $\frac{1}{3}$

D. $\frac{1}{2}$

E. $\frac{2}{3}$



9. If p, q, r, and s are as shown on the number line above, which of the following products is greatest?

A. pq

B. pr

C. qr

D. qs

E. rs

10. Last year $48,000 of a certain store's profit was shared by its 2 owners and their 10 employees. Each of the 2 owners received 3 times as much as each of their 10 employees. How much did each owner receive from the $48,000?

A. $12,000

B. $9,000

C. $6,000

D. $4,000

E. $3,000

11. On a vacation, Rose exchanged $500.00 for euros at an exchange rate of 0.80 euro per dollar and spent $\frac{3}{4}$ of the euros she received. If she exchanged the remaining euros for dollars at an exchange rate of $1.20 per euro, what was the dollar amount she received?

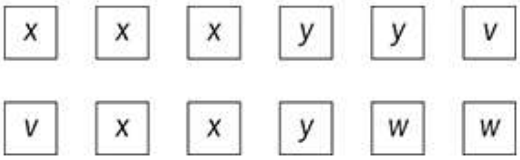
A. $60.00

B. $80.00

C. $100.00

D. $120.00

E. $140.00



12. Each of the 12 squares shown is labeled x, y, v, or w. What is the ratio of the number of these squares labeled x or y to the number of these squares labeled v or w?

A. 1:2

B. 2:3

C. 4:3

D. 3:2

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E. 2:1

Number of Solid-Colored Marbles in Three Jars			
Jar	Number of red marbles	Number of green marbles	Total number of red and green marbles
P	x	y	80
Q	y	z	120
R	x	z	160

13. In the table above, what is the number of green marbles in Jar R?

A. 70

B. 80

C. 90

D. 100

E. 110

14. Bouquets are to be made using white tulips and red tulips, and the ratio of the number of white tulips to the number of red tulips is to be the same in each bouquet. If there are 15 white tulips and 85 red tulips available for the bouquets, what is the greatest number of bouquets that can be made using all the tulips available?

A. 3

B. 5

C. 8

D. 10

E. 13

County	Amount Recycled	Amount Disposed of
A	16,700	142,800
B	8,800	48,000
C	13,000	51,400
D	3,900	20,300
E	3,300	16,200

15. The table above shows the amount of waste material, in tons, recycled by each of five counties in a single year and the amount of waste material, also in tons, that was disposed of in landfills by the five counties in that year. Which county had the lowest ratio of waste material disposed of to waste material recycled in the year reported in the table?

A. A

B. B

C. C

D. D

E. E

16. 125% of 5 =

A. 5.125

B. 5.25

C. 6

D. 6.125

E. 6.25

17. Each day Linda spends 6 hours at her dog-walking job, for which she is paid $15.00 per hour. In addition, Linda

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17. Each day Linda spends 6 hours at her dog-walking job, for which she is paid $15.00 per hour. In addition, Linda spends a total of 1.5 hours each day, (for which she is not paid), traveling to and from the job. What is Linda's hourly rate of pay for the total amount of time that she spends each day at her dog-walking job and traveling to and from that job?

    A. $7.50

    B. $9.00

    C. $10.00

    D. $10.75

    E. $12.00

18. When traveling at a constant speed of 32 miles per hour, a certain motorboat consumes 24 gallons of fuel per hour. What is the fuel consumption of this boat at this speed measured in miles traveled per gallon of fuel?

    A. $\frac{2}{3}$

    B. $\frac{3}{4}$

    C. $\frac{4}{5}$

    D. $\frac{4}{3}$

    E. $\frac{3}{2}$

19. If snow accumulation increased at a constant rate of 30 millimeters per hour during a certain snowstorm, how many <u>seconds</u> did it take for snow accumulation to increase by 1 millimeter?

    A. $\frac{1}{120}$

    B. $\frac{1}{60}$

    C. $\frac{1}{20}$

    D. 20

    E. 120

20. From 2000 to 2003, the number of employees at a certain company increased by a factor of $\frac{1}{4}$. From 2003 to 2006, the number of employees at this company decreased by a factor of $\frac{1}{3}$. If there were 100 employees at the company in 2006, how many employees were there at the company in 2000?

    A. 200

    B. 120

    C. 100

    D. 75

    E. 60

21. Working alone at its own constant rate, machine R fills 10,000 boxes with pins in 9 hours, and working alone at its own constant rate, machine S fills 5,000 boxes with pins in 3 hours. If machine R and machine S, each working at its own constant rate and for the same period of time, together filled a certain number of boxes with pins, what percent of the boxes were filled by machine R?

    A. 25%

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A. 25%

B. 33%

C. 40%

D. 66%

E. 75%

22. A collection of 16 coins, each with a face value of either 10 cents or 25 cents, has a total face value of $2.35. How many of the coins have a face value of 25 cents?

A. 3

B. 5

C. 7

D. 9

E. 11

23. A retailer purchased eggs at $2.80 per dozen and sold the eggs at 3 eggs for $0.90. What was the retailer's gross profit from purchasing and selling 5 dozen eggs? (1 dozen eggs = 12 eggs)

A. $0.90

B. $2.40

C. $4.00

D. $11.30

E. $12.00

24. In a set of 24 cards, each card is numbered with a different positive integer from 1 to 24. One card will be drawn at random from the set. What is the probability that the card drawn will have either a number that is divisible by both 2 and 3 or a number that is divisible by 7?

A. $\frac{3}{24}$

B. $\frac{4}{24}$

C. $\frac{7}{24}$

D. $\frac{8}{24}$

E. $\frac{17}{24}$

25. As a salesperson, Phyllis can choose one of two methods of annual payment: either an annual salary of $35,000 with no commission or an annual salary of $10,000 plus a 20% commission on her total annual sales. What must her total annual sales be to give her the same annual pay with either method?

A. $100,000

B. $120,000

C. $125,000

D. $130,000

E. $132,000

26. If $1 < x < y < z$, which of the following has the greatest value?

A. $z(x+1)$

B. $z(y+1)$

C. $x(y+z)$

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C. $x(y+z)$

D. $y(x+z)$

E. $z(x+y)$

27. Set X consists of eight consecutive integers. Set Y consists of all the integers that result from adding 4 to each of the integers in Set X and all the integers that result from subtracting 4 from each of the integers in Set X. How many more integers are there in Set Y than in Set X?

A. 0

B. 4

C. 8

D. 12

E. 16

28. Of the following, which is the closest to $\frac{60.2}{1.03 \times 4.86}$?

A. 10

B. 12

C. 13

D. 14

E. 15

29. Thabo owns exactly 140 books, and each book is either paperback fiction, paperback nonfiction, or hardcover nonfiction. If he owns 20 more paperback nonfiction books than hardcover nonfiction books, and twice as many paperback fiction books as paperback nonfiction books, how many hardcover nonfiction books does Thabo own?

A. 10

B. 20

C. 30

D. 40

E. 50

30. What is 35 percent of the sum of 1.4 and $\frac{1}{5}$?

A. 0.42

B. 0.56

C. 0.85

D. 1.55

E. 1.95

31. If $x > 0$, $\frac{x}{50} + \frac{x}{25}$ is what percent of x?

A. 6%

B. 25%

C. 37%

D. 60%

E. 75%

32. During a certain time period, Car X traveled north along a straight road at a constant rate of 1 mile per minute and used fuel at a constant rate of 5 gallons every 2 hours. During this time period, if Car X used exactly 3.75

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gallons of fuel, how many miles did Car X travel?

A. 36

B. 37.5

C. 40

D. 80

E. 90

33. Which of the following is closest to the value of $\dfrac{999}{100+\dfrac{1}{999}}$

A. 10

B. 1

C. 0.1

D. 0.01

E. 0.001

34. The population of City X is 50 percent of the population of City Y. The population of City X is what percent of the total population of City X and City Y?

A. 25%

B. $33\frac{1}{3}\%$

C. 40%

D. 50%

E. $66\frac{2}{3}\%$

35. A bicycle store purchased two bicycles, one for $250 and the other for $375, and sold both bicycles at a total gross profit of $250. If the store sold one of the bicycles for $450, which of the following could be the store's gross profit from the sale of the other bicycle?

A. $75

B. $100

C. $125

D. $150

E. $175

36. If $k^2 = m^2$, which of the following must be true?

A. $k = m$

B. $k = -m$

C. $k = |m|$

D. $k = -|m|$

E. $|k| = |m|$

37. Makoto, Nishi, and Ozuro were paid a total of $780 for waxing the floors at their school. Each was paid in proportion to the number of hours he or she worked. If Makoto worked 15 hours, Nishi worked 20 hours, and Ozuro worked 30 hours, how much was Makoto paid?

A. $52

B. $117

C. $120

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C. $130

D. $180

E. $234

38. If x is a positive integer and $4^x - 3 = y$, which of the following CANNOT be a value of y?

A. 1

B. 7

C. 13

D. 61

E. 253

39. The regular price per can of a certain brand of soda is $0.40. If the regular price per can is discounted 15 percent when the soda is purchased in 24-can cases, what is the price of 72 cans of this brand of soda purchased in 24-can cases?

A. $16.32

B. $18.00

C. $21.60

D. $24.48

E. $28.80

40. The quotient when a certain number is divided by $\frac{2}{3}$ is $\frac{9}{2}$. What is the number?

A. $\frac{4}{27}$

B. $\frac{1}{3}$

C. 3

D. 6

E. $\frac{27}{4}$

41. The price of gasoline at a service station increased from $1.65 per gallon last week to $1.82 per gallon this week. Sally paid $26.40 for gasoline last week at the station. How much more will Sally pay this week at the station for the same amount of gasoline?

A. $1.70

B. $2.55

C. $2.64

D. $2.72

E. $2.90

42. Harry spent 25% of his $2,200 May take-home pay on rent and 30% of the remainder on food. How much of his May take-home pay did Harry spend on food?

A. $165

B. $495

C. $550

D. $660

E. $1,045

43. If $2x + y = 7$ and $x + 2y = 5$, then $x + y =$

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43. If $2x + y = 7$ and $x + 2y = 5$, then $\frac{[illegible]}{3} =$

A. 1

B. $\frac{4}{3}$

C. $\frac{17}{5}$

D. $\frac{18}{5}$

E. 4

44. City X has a population 4 times as great as the population of City Y, which has a population twice as great as the population of City Z. What is the ratio of the population of City X to the population of City Z?

A. 1:8

B. 1:4

C. 2:1

D. 4:1

E. 8:1

Total Membership of Organization X, 1988

Emeritus	78
Fellows	9,209
Partners	35,509
Associates	27,909
Affiliates	2,372

45. According to the table above, the number of fellows was approximately what percent of the total membership of Organization X? **Note: All categories of membership are mutually exclusive.**

A. 9%

B. 12%

C. 18%

D. 25%

E. 35%

46. If $S = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \frac{1}{6^2} + \frac{1}{7^2} + \frac{1}{8^2} + \frac{1}{9^2} + \frac{1}{10^2}$, which of the following is true?

A. $S > 3$

B. $S = 3$

C. $2 < S < 3$

D. $S = 2$

E. $S < 2$

47. A manufacturer of a certain product can expect that between 0.3 percent and 0.5 percent of the units manufactured will be defective. If the retail price is $2,500 per unit and the manufacturer offers a full refund for defective units, how much money can the manufacturer expect to need to cover the refunds on 20,000 units?

A. Between $15,000 and $25,000

B. Between $30,000 and $50,000

C. Between $60,000 and $100,000

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D. Between \$150,000 and \$250,000

E. Between \$300,000 and \$500,000

48. Which of the following is closest to $\frac{\sqrt{4.2(1{,}590)}}{15.7}$?

A. 20

B. 40

C. 60

D. 80

E. 100

49. What is the value of $\sqrt{17}+\sqrt{47}$, to the nearest whole number?

A. 8

B. 11

C. 12

D. 15

E. 32

50. Last week Chris earned x dollars per hour for the first 40 hours worked plus 22 dollars per hour for each hour worked beyond 40 hours. If last week Chris earned a total of 816 dollars by working 48 hours, what is the value of x?

A. 13

B. 14

C. 15

D. 16

E. 17

51. The value of $\frac{\frac{7}{8}+\frac{1}{9}}{\frac{1}{2}}$ is closest to which of the following?

A. 2

B. $\frac{3}{2}$

C. 1

D. $\frac{1}{2}$

E. 0

52. The positive two-digit integers x and y have the same digits, but in reverse order. Which of the following must be a factor of $x+y$?

A. 6

B. 9

C. 10

D. 11

E. 14

53. In a certain sequence of 8 numbers, each number after the first is 1 more than the previous number. If the first

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53. In a certain sequence of 8 numbers, each number after the first is 1 more than the previous number. If the first number is –5, how many of the numbers in the sequence are positive?

A. None

B. One

C. Two

D. Three

E. Four

54. A total of s oranges are to be packaged in boxes that will hold r oranges each, with no oranges left over. When n of these boxes have been completely filled, what is the number of boxes that remain to be filled?

A. $s-nr$

B. $s-\frac{n}{r}$

C. $rs-n$

D. $\frac{s}{n}-r$

E. $\frac{s}{r}-n$

55. If $0<a<b<c$, which of the following statements must be true?

I. $2a>b+c$

II. $c-a>b-a$

III. $\frac{c}{a}<\frac{b}{a}$

A. I only

B. II only

C. III only

D. I and II

E. II and III

	Monday	Tuesday	Wednesday	Thursday
Company A	45	55	50	50
Company B	10	30	30	10
Company C	34	28	28	30
Company D	39	42	41	38
Company E	50	60	60	70

56. The table shows the numbers of packages shipped daily by each of five companies during a 4-day period. The standard deviation of the numbers of packages shipped daily during the period was greatest for which of the five companies?

A. A

B. B

C. C

D. D

E. E

57. Company Q plans to make a new product next year and sell each unit of this new product at a selling price of $2. The variable costs per unit in each production run are estimated to be 40% of the selling price, and the fixed costs for each production run are estimated to be $5,040. Based on these estimated costs, how many units of the

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costs for each production run are estimated to be $5,040. Based on these estimated costs, how many units of the new product will Company Q need to make and sell in order for their revenue to equal their total costs for each production run?

A. 4,200

B. 3,150

C. 2,520

D. 2,100

E. 1,800

58. A small business invests $9,900 in equipment to produce a product. Each unit of the product costs $0.65 to produce and is sold for $1.20. How many units of the product must be sold before the revenue received equals the total expense of production, including the initial investment in equipment?

A. 12,000

B. 14,500

C. 15,230

D. 18,000

E. 20,000

Estimated Number of Home-Schooled Students by State, January 2001

State	Number (in thousands)
A	181
B	125
C	103
D	79
E	72

59. According to the table shown, the estimated number of home-schooled students in State A is approximately what percent greater than the number in State D?

A. 25%

B. 55%

C. 100%

D. 125%

E. 155%

60. When n liters of fuel were added to a tank that was already $\frac{1}{3}$ full, the tank was filled to $\frac{7}{9}$ of its capacity. In terms of n, what is the capacity of the tank, in liters?

A. $\frac{10}{9}n$

B. $\frac{4}{3}n$

C. $\frac{3}{2}n$

D. $\frac{9}{4}n$

E. $\frac{7}{3}n$

61. A bicycle race on a mountain trail began at an elevation 850 meters above Town X. The race course consisted of

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61. A bicycle race on a mountain trail began at an elevation 850 meters above Town X. The race course consisted of 5 phases, with the following elevation changes: a descent of 280 meters, an ascent of 350 meters, a descent of 620 meters, an ascent of 100 meters, and a descent of 400 meters. At what elevation, in meters above or below Town X, did the race end?

    A. 850 below Town X

    B. Same elevation as Town X

    C. 850 above Town X

    D. 1,700 above Town X

    E. 2,600 above Town X

62. The harvest yield from a certain apple orchard was 350 bushels of apples. If x of the trees in the orchard each yielded 10 bushels of apples, what fraction of the harvest yield was from these x trees?

    A. $\frac{x}{35}$

    B. $1 - \frac{x}{35}$

    C. $10x$

    D. $35 - x$

    E. $350 - 10x$

63. In a certain fraction, the denominator is 16 greater than the numerator. If the fraction is equivalent to 80 percent, what is the denominator of the fraction?

    A. 32

    B. 64

    C. 72

    D. 80

    E. 120

64. Jonathan, Kyle, and Michael agreed to pay equal shares of the monthly rent on an apartment. Because Jonathan has a cat, he paid a 1-time pet deposit of $250. The pet deposit and Jonathan's share of the monthly rent for the first month totaled $525. How much is the monthly rent on the apartment?

    A. $1,575

    B. $1,075

    C. $1,025

    D. $825

    E. $775

65.

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